

SPY 1-Month Risk-Reversal Trade Analysis

A Quantitative Evaluation of Skew, Volatility Regimes, and Alternative Structures

Rhugved Bhojane

November 2025

Abstract

This report develops a complete quantitative framework for evaluating a 1-month SPY risk-reversal strategy as of April 9, 2025. The analysis integrates a Black–Scholes–Merton pricing engine, scenario-based P&L modeling, and a two-year historical study of realized volatility, implied volatility, volatility risk premium (VRP), and risk-reversal skew. The performance of the baseline risk-reversal is compared with two alternative structures: a put-spread enhancement and buying SPY outright. The study provides a full view of the trade’s convexity profile, downside tail risk, and sensitivity to prevailing volatility regimes.

1 Introduction

This project investigates whether a 1-month SPY risk-reversal should be entered as of April 9, 2025. The structure consists of selling a 95% moneyness put and buying a 105% moneyness call, each executed with 1,000 option contracts. This structure expresses a bullish, skew-selling view: the trader sells expensive downside implied volatility and finances the purchase of cheap upside convexity.

To support this assessment, the project builds a modular Python-based system including:

- A robust data ingestion pipeline for SPY prices, SOFR rates, dividend yields, and implied volatilities.
- A Black–Scholes–Merton pricing engine for valuing options and computing mark-to-market P&L.
- A scenario engine for evaluating P&L one week prior to expiry.
- A historical volatility analysis covering realized volatility, ATM implied volatility, VRP, and risk-reversal skew.
- Bonus evaluation of a put-spread-enhanced structure and SPY outright exposure.

2 Data Description

Three datasets were provided:

1. **Index Data.xlsx**: SPY spot, VIX close, SOFR close, dividend amounts, and implied dividend yields.
2. **IV.xlsx**: 1-month SPY implied volatilities for ATM, 95% (downside), and 105% (upside) moneyness.
3. **options_parsed.xlsx**: Option chain containing mid-price, delta, gamma, vega, theta, and implied volatility for strikes surrounding the trade.

All data spans January 2023–April 2025, enabling robust historical analysis.

Key extracted inputs for April 9, 2025:

- Spot: $S_0 = 504.38$
- Risk-free rate (SOFR): $r = 4.42\%$
- Dividend yield: $q = 1.34\%$
- Put strike: $K_{\text{put}} \approx 480$
- Call strike: $K_{\text{call}} \approx 530$
- Volatilities:

$$\sigma_{95\%} \approx 33.9\%, \quad \sigma_{105\%} \approx 18.0\%$$

3 Methodology

The analysis proceeds in five steps:

1. Extract market inputs for the trade date.
2. Price the put and call legs at inception using Black–Scholes–Merton.
3. Simulate mark-to-market valuations for spot moves of -15% to $+15\%$ one week prior to expiry.
4. Compute historical realized volatility, ATM implied volatility, VRP, and skew.
5. Evaluate alternative structures (put-spread and SPY outright).

Time to maturity is computed using $\text{ACT}/365$. Volatility is assumed constant in mark-to-market scenarios.

4 Option Pricing Model

We use the Black–Scholes–Merton model with continuous dividend yield q :

$$d_1 = \frac{\ln(S/K) + (r - q + 0.5\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

Call and put prices:

$$C = Se^{-qT}N(d_1) - Ke^{-rT}N(d_2),$$

$$P = Ke^{-rT}N(-d_2) - Se^{-qT}N(-d_1).$$

Model outputs included entry prices, deltas, vegas, and scenario valuations.

5 Scenario P&L One Week Before Expiry

We evaluate the structure under seven spot scenarios from -15% to $+15\%$. The mark-to-market P&L results are summarized in Table 1.

Scenario	Spot	P&L (USD)
-15%	428.72	-4.48M
-10%	453.94	-2.08M
-5%	479.16	-0.30M
0%	504.38	+0.47M
+5%	529.60	+1.14M
+10%	554.82	+3.16M
+15%	580.04	+5.66M

Table 1: Risk-Reversal P&L One Week Prior to Expiry

The asymmetry is clear:

- Upside convexity produces strong profits at large positive moves.
- Downside convexity generates large losses due to the naked short put.

6 Historical Volatility and Skew Analysis

We compute:

- 21-day realized volatility,
- 1-month ATM implied volatility,
- Volatility risk premium ($VRP = IV - RV$),
- 95–105% risk-reversal skew.

Summary statistics appear in Table 2.

Statistic	Realized Vol	ATM IV	VRP	Skew
Mean	13.35%	14.07%	0.73%	6.68%
Std. Dev	5.50%	3.95%	3.93%	1.49%
Min	5.76%	9.33%	-24.9%	2.53%
Median	12.31%	12.95%	0.93%	6.56%
Max	50.69%	42.32%	11.1%	15.85%

Table 2: Historical Volatility and Skew Summary

Key regime findings for April 9, 2025:

- VRP is at the **0.5th percentile**: implied volatility is extremely cheap.
- Skew is at the **99.8th percentile**: downside puts are extraordinarily expensive relative to upside calls.

This creates a conflict: skew strongly favors selling downside volatility, but low VRP makes naked short-vol trades dangerous.

7 Bonus (a): Put-Spread + Call Structure

To reduce downside tail risk, we evaluate a modified structure:

- Short 95% put,
- Long 90% put,
- Long 105% call.

Scenario results:

- 15% downside loss improves from **-4.48M** to **-2.14M**.
- Upside outcome remains strong at **+5.27M** vs +5.66M.

Interpretation:

The put-spread preserves skew-selling benefits while capping tail risk. Given the combination of rich skew and low VRP, the put-spread offers a superior risk-reward trade-off compared to the naked risk-reversal.

8 Bonus (b): Comparison with Buying SPY Outright

SPY outright provides linear delta exposure:

$$\Delta P = N(S_T - S_0).$$

Scenario results reveal:

- Largest upside in strong bull markets,
- Largest downside losses,
- No convexity or skew exposure.

SPY outright is preferable when:

- The market is in a strong trending bull regime,
- Skew is normal or cheap,
- Implied volatility is expensive relative to realized,
- The investor prefers delta exposure and avoids short-volatility risk.

9 Conclusion

This project delivers a complete quantitative assessment of SPY risk-reversal structures under prevailing market conditions. We find that the trade date of April 9, 2025 exhibits a highly unusual combination of:

- Extremely rich downside skew (99.8th percentile),
- Extremely cheap implied volatility relative to realized (0.5th percentile).

This environment makes selling skew attractive but selling volatility dangerous. The baseline risk-reversal carries large downside convexity and substantial tail risk. The put-spread enhancement significantly reduces these risks while preserving the relative-value benefits of rich skew.

SPY outright becomes optimal only in strongly bullish markets with expensive IV and subdued skew.

Overall, the put-spread + call structure provides the best balance of directional exposure, skew monetization, and controlled downside risk for the observed market regime.